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# =====  
# Meter Remark Classification AI  
# Logistic Regression + TF-IDF  
# =====  
  
# 1 Import Libraries  
import pandas as pd  
import pickle  
  
from sklearn.model_selection import train_test_split  
from sklearn.feature_extraction.text import TfidfVectorizer  
from sklearn.linear_model import LogisticRegression  
from sklearn.metrics import accuracy_score, classification_report  
  
# 2 Load Dataset  
data = pd.read_csv("clean_meter_data.csv")  
  
print("Dataset Loaded Successfully")  
print(data.head())  
  
# 3 Define Input and Output  
X = data['TEXT_REMARKS']  
y = data['ACTION_CATEGORY']  
  
# 4 TF-IDF Feature Extraction  
tfidf = TfidfVectorizer()  
  
X_tfidf = tfidf.fit_transform(X)  
  
print("TF-IDF Vectorization Completed")  
  
# 5 Train-Test Split  
X_train, X_test, y_train, y_test = train_test_split(  
    X_tfidf,  
    y,  
    test_size=0.2,  
    random_state=42  
)  
  
print("Dataset Split Completed")  
  
# 6 Train Logistic Regression Model  
model = LogisticRegression(max_iter=1000)  
  
model.fit(X_train, y_train)  
  
print("Model Training Completed")  
  
# 7 Model Evaluation  
y_pred = model.predict(X_test)  
  
print("\nModel Accuracy:", accuracy_score(y_test, y_pred))  
  
print("\nClassification Report:")  
print(classification_report(y_test, y_pred))  
  
# 8 Save Model and TF-IDF  
pickle.dump(model, open("meter_remark_model.pkl", "wb"))  
pickle.dump(tfidf, open("tfidf_vectorizer.pkl", "wb"))  
  
print("\nModel Saved Successfully")  
  
# 9 Prediction Function  
def predict_department(text):  
  
    text_vec = tfidf.transform([text])  
  
    prediction = model.predict(text_vec)
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    return prediction[0]

# 10 User Input System
print("\n===== Meter Remark Classification System =====")

while True:

    remark = input("\nEnter Meter Testing Remark (or type exit): ")

    if remark.lower() == "exit":
        break

    result = predict_department(remark)

    print("Concerned Department:", result)
```

Dataset Loaded Successfully

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TEXT_REMARKS \
0 Mentioned meter is not found at mentioned addr...
1 Meter not found on Specified location also wro...
2 Address Not found At the Mentioned pole So Met...
3 Address Not found At the Mention Pole Mobile n...
4 No responsible person found at site also no re...
```

ACTION_CATEGORY

```
0 Field Operations / Meter Testing
1 Field Operations / Meter Testing
2 Field Operations / Meter Testing
3 Field Operations / Meter Testing
4 Field Operations / Meter Testing
TF-IDF Vectorization Completed
Dataset Split Completed
Model Training Completed
```

Model Accuracy: 0.9938808373590983

Classification Report:

	precision	recall	f1-score	support
Back-Office / Administrative Operations	0.94	0.92	0.93	36
CEG / Enforcement / Vigilance Team	1.00	0.97	0.98	33
Customer Service / Call Center	1.00	0.97	0.99	36
Field Operations / Meter Testing	0.95	0.97	0.96	89
Meter Replacement / Installation Team	0.98	1.00	0.99	320
Network / Zone Maintenance Team	1.00	0.38	0.55	8
No Action Needed	1.00	1.00	1.00	2558
Scheduling / Field Dispatch Team	0.96	0.96	0.96	25
accuracy			0.99	3105
macro avg	0.98	0.89	0.92	3105
weighted avg	0.99	0.99	0.99	3105

Model Saved Successfully

===== Meter Remark Classification System =====

Enter Meter Testing Remark (or type exit): Meter Testing Done and found OK
Concerned Department: No Action Needed

Enter Meter Testing Remark (or type exit): Meter Display Found Defective and Replacemnet Reuired for same, also Seal is missing
Concerned Department: Meter Replacement / Installation Team

Enter Meter Testing Remark (or type exit): Seal is missing
Concerned Department: CEG / Enforcement / Vigilance Team

Enter Meter Testing Remark (or type exit): Tampered Meter Found at site
Concerned Department: Field Operations / Meter Testing

Enter Meter Testing Remark (or type exit): Meter tampered
Concerned Department: No Action Needed

Enter Meter Testing Remark (or type exit): Tamper
Concerned Department: No Action Needed

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